a fundamentally different situation from a regular call option, which pays out a dollar for each dollar the asset price finishes above the strike. Think about it: suppose we are short this binary call, it expires tomorrow, and the underlying asset is currently at \$49.99. If the underlying finishes at \$49.99, we owe nothing, but if it finishes at \$50.00 we owe \$1mm. Imagine we are short this option with one day to go and the underlying is near-the-money. How are we going to hedge this? We're not. It's a complete nightmare. Our delta explodes if we are near-the-money; the amount of the underlying that we would need to buy or sell as the underlying oscillates near the strike is enormous. Transaction costs associated with trading the underlying—not to mention the effect that one may have on the price of the underlying itself through trying to trade such large size in the underlying in order to hedge—would be significant.[41]

A.7.2 Vega of Binary Options

Let's look now at the vega of binary options. As we discussed in Section A.6.3, the vega of an option tells us how much the value of the option will change when implied volatility changes. This is an item of primary importance to an option trader, whose job is often to manage vega risk and/or take views on movements in implied volatility in the options markets.

Under the Black-Scholes assumptions, the vega of a cash-or-nothing binary call is as follows:

$$\begin{aligned}\frac{\partial c_{CN}}{\partial \sigma} &= \frac{\partial \left[Ke^{-rT}N(d_2)\right]}{\partial \sigma}\\ &= -Ke^{-rT}n(d_2)\left[\frac{\ln(S/Xe^{-rT})}{\sigma^2\sqrt{T}} + 0.5\sqrt{T}\right]\\ &= -Ke^{-rT}n(d_2)\frac{d_1}{\sigma}.\end{aligned}$$

Similarly, it can be shown that under the Black-Scholes assumptions the vega of a cash-or-nothing binary put is

$$\frac{\partial p_{CN}}{\partial \sigma} = Ke^{-rT}N(d_2)\frac{d_1}{\sigma}.$$

[41]Indeed, an option trader who is short a binary call that is near-the-money may try to prevent the option from finishing in-the-money by selling the underlying itself, which exerts downward pressure on the price (the legality of this is somewhat of a gray area, depending in part on the particular marketplace in which the underlying trades). If the option trader sells enough of the underlying, he may be able to drive the price down to prevent the option from finishing in-the-money. Of course, other market forces may be at play that could make this difficult.

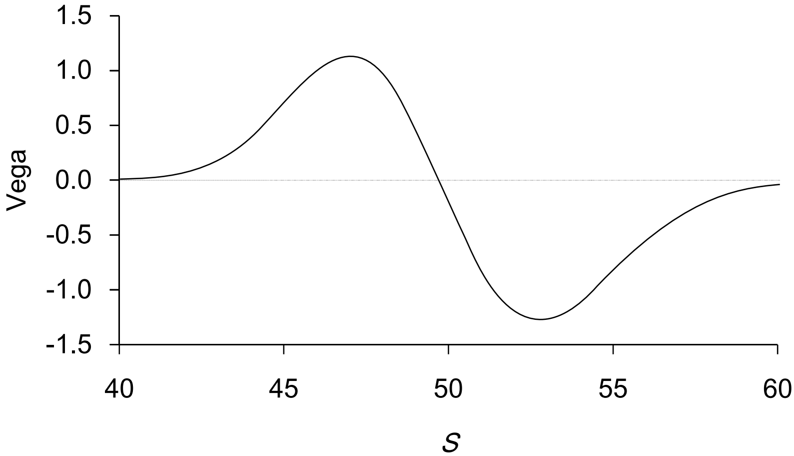

Figure A.22: Vega of a Cash-or-Nothing Binary Call as a Function of Underlying Asset When $K = 1$, $X = 50$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ Month

If looking at the computation of the analytical expression for the vega of the option is unsettling, one could always resort to approximating the vega computationally, as in Equation A.51.

Figure A.22 shows the vega of a cash-or-nothing call as a function of the underlying asset. This vega profile is very unusual as compared to ordinary options. When the option is in-the-money, the holder of the option is *short vol*, meaning that an increase in volatility actually decreases the value of the option. When the option is out-of-the-money, the holder of the option is typically *long vol*, meaning that an increase in implied volatility increases the value of the option.[42] What's going on here? Once the option is in-the-money, that's it: life is not going to get any better. The maximum payout from the option is K. There is no further upside. When the option is in-the-money, the holder wants nothing more than for volatility to go to zero, for the underlying to stop moving around, so that he can get that money K. When the option is far enough out-of-the-money, the holder is long vol because he needs the underlying to move enough so that the option finishes in-the-money. So an option trader who trades in this option may start out with a certain vol position, only to find that his vol position completely flips around as the underlying moves.

[42] As seen in Figure A.22, the sign of the vega does not change when the underlying is equal to the strike, but rather slightly below the strike.

A.8 Packages

Packages refer to options that are formed from a combination of other options, cash, and the underlying asset itself. The value of a package must be equal to the sum of the prices of the component parts, as dictated by arbitrage.

Being able to recognize an option as a package of other building blocks is a very important skill to master. It usually requires staring at a picture of the payout of the option and trying to figure out how to break the picture down into more simple components. Oftentimes while doing so, one makes some wrong turns and has to go back and try again. But you do enough of these and you get the hang of it. We use this technique, for example, deconstructing the knockout swap in Chapter 6.

Example

An *asset-or-nothing option* provides the holder with either nothing or a number of units of the underlying asset. Consider an asset-or-nothing call that provides the holder with one unit of the underlying asset if the underlying

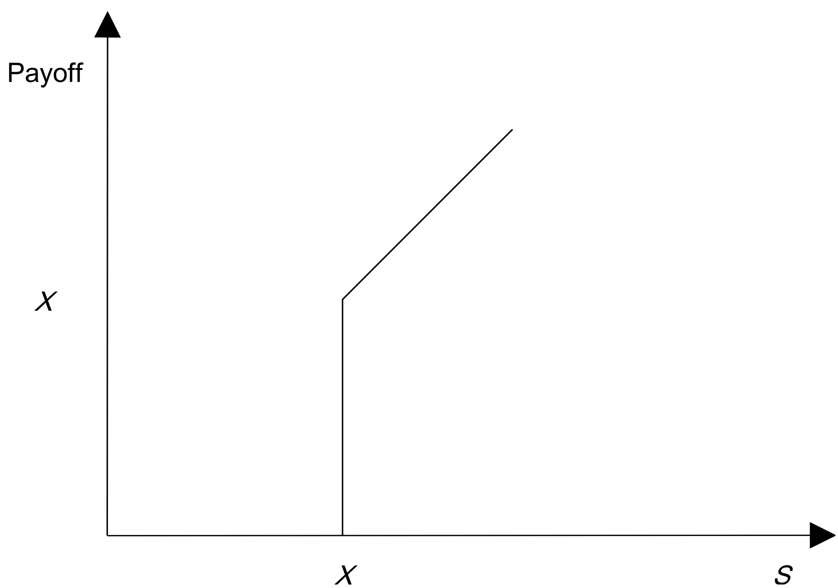

Figure A.23: Payoff of an Asset-or-Nothing Call is above the strike and zero otherwise. The payoff from the option is:

$$\text{Payoff} = \begin{cases} S_T & \text{if } S_T \geq X \\ 0 & \text{if } S_T < X. \end{cases}$$

This payoff pattern is shown in Figure A.23. If you stare at Figures A.16 and A.23 long enough, perhaps you will recognize that the payoff of an asset-or-nothing call is the same as the payoff of a portfolio consisting of an ordinary call option on the underlying struck at X and a cash-or-nothing binary call struck at X with a payoff K set equal to X.[43] Since both alternatives offer the same payoffs, they must be worth the same today.[44] Thus,

$$\begin{aligned} C_{AN} &= c(X) + c_{CN}(X) \\ &= SN(d_1) - Xe^{-rT}N(d_2) + Xe^{-rT}N(d_2) \\ &= SN(d_1). \end{aligned}$$

□

[43] Let's show this is the case, by looking at the payoff of the portfolio at expiration. If $S_T < X$, then both the call option and the cash-or-nothing binary call option pay nothing, in which case the portfolio pays nothing. If, however, $S_T \geq X$, then the call option pays out $S_T - X$ and the binary pays out X, for a total of $S_T - X + X = S_T$. Thus, this portfolio has the same payout as the asset-or-nothing binary call.

[44] If they were not worth the same, you could arbitrage this situation by buying the cheaper one and selling the more expensive one, which gives you positive cash flow today and zero cash flow in the future.

Supplemental Problem Set

1. Consider a stock that pays no dividends and is expected to be particularly volatile over the coming two periods. The current price of the stock is \$32 per share, and the binomial tree in Figure A.24 describes the possible stock prices over the next two periods.

European calls and puts trade on the stock. Consider a European call with a strike of \$40 and a European put with a strike of \$40, both of which expire in two periods. Assume that the risk-free interest rate each period is $33\frac{1}{3}\%$.

a) What is the value today of the European call with a strike of \$40? What is the appropriate replicating portfolio of underlying stock and bonds at each node?

b) What is the value today of the European put with a strike of \$40? What is the appropriate replicating portfolio of underlying stock and bonds at each node?

c) Verify that these two prices satisfy European put-call parity.

d) What is the price today of an American call option with a strike of \$40?

e) What is the price today of an American put option with a strike of \$40?

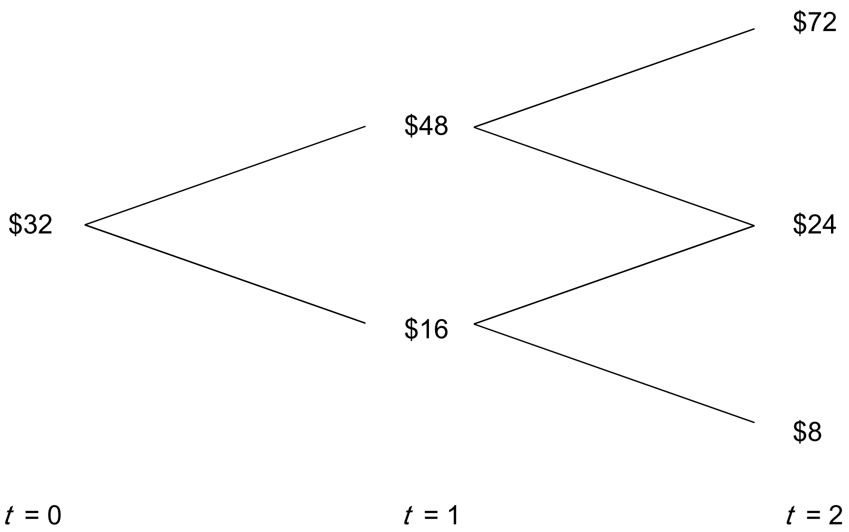

Figure A.24: A Two-Period Binomial Model of Stock Price

2. Consider a compound option on the stock from the first problem (i.e., use the binomial tree in Figure A.24). This compound option is a put on a call. At time $t = 1$, the holder of the compound option has the right to receive the first strike price, $X_1 = \$20$, and deliver a call option. This call option that will be delivered in the event of exercise expires at time $t = 2$ and has a strike of $X_2 = \$15$. What is the value of the compound option today? Assume the risk-free rate each period in $33\frac{1}{3}\%$.

3. Consider a stock that pays no dividends and whose current price is \$40. The binomial tree in Figure A.25 describes the possible stock prices over the next several periods.

Assume that the risk-free rate each period is 50%.

a) Consider a European lookback put option on the stock. The payoff at maturity of the option is equal to $\max(S_{\max} - S_T, 0)$, where $S_{\max}$ is the maximum stock price reached during the life of the option, and S_T is the final stock price. What is the value of the European lookback put option?

b) Now consider an American lookback put option on the stock. If exercised at any time t prior to maturity, the payoff is equal to the difference between the maximum stock price that occurred between time 0 and time t and the current stock price. What is the value of the American lookback put option?

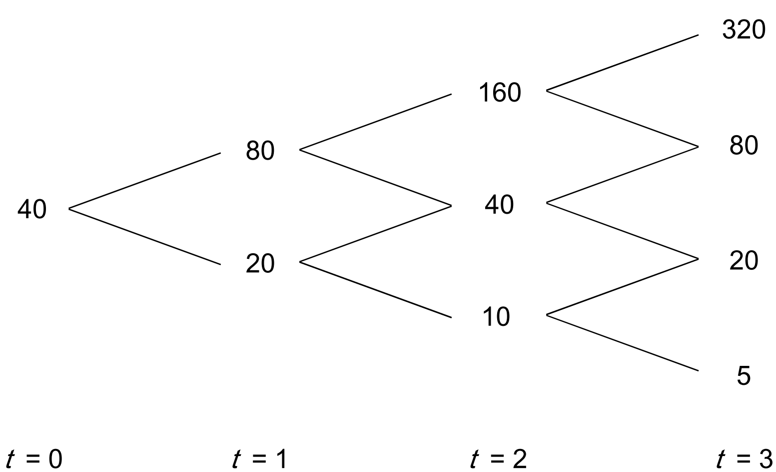

Figure A.25: Binomial Model of Stock Price

4. Consider a company that is a clothing manufacturer who is having financial trouble. Their stock does not pay any dividends, and the price is currently \$100. Assume that the stock price will either be \$120 or \$20 at the end of one year. In an effort to raise cash today, the company is running the following promotion: with the purchase of every suit, a customer will receive the choice of a) \$8 cash today or b) a contract that specifies that if the company's stock price at the end of one year is above its current level of \$100, the customer will receive \$20 one year from today. If, however, the stock price is below \$100, the customer will receive nothing.

Assume that the company's stock is actively traded and that the risk-free rate is 8%. Assuming you buy a suit from the company, which would you choose (i.e., which financial asset do you choose, not which suit)?

5. Consider a stock whose current price is \$20. The distribution of the security is lognormal. Assume that the continuously compounded risk-free rate is 10% per annum and denote by σ the volatility of the stock per year.

Consider a derivative contract written on the stock. This contract will pay \$1 if the price of the stock is at least \$25 in one year. This contract is worth \$0.34 today. What is the value today of a contract that will pay off \$1 if the price of the stock is less than \$25 in one year?

6. A European put option with one month until expiration is selling for \$2.50. The stock pays no dividends. The current stock price is \$47, the strike is \$50, and the continuously compounded risk-free rate is 6%. Is there an arbitrage available?

7. Let $c_1, c_2,$ and c_3 be the prices of European call options with strikes $X_1, X_2,$ and X_3, respectively. Suppose that $X_3 > X_2 > X_1$, where $X_3 - X_2 = X_2 - X_1$. All options are on the same stock and have the same maturity. Show that c_2 is less than or equal to $(c_1 + c_3)/2$.

8. Consider an option on a stock. Suppose the stock price is currently \$30 and assume that the stock price is lognormally distributed. Assume that the exercise price is \$29, the continuously compounded risk-free rate is 5%, the volatility of the stock price is 25%, and the time to maturity of the option is four months. Find the price of the option if it is a) a European call, b) an American call, or c) a European put. Show that put-call parity holds.

9. A European put with six months maturity is selling for \$6.50. The strike price of the put is \$40, and the underlying stock price is currently \$32.

The continuously compounded risk-free rate is 2%. Is there an arbitrage opportunity?

10. Describe a portfolio of options that is long gamma and short vega. (You don't need to give any numbers here. For example, if the question asked you to describe a portfolio of options that was long just gamma, an appropriate response would be "long call options.")

11. Suppose ABC and XYZ are two non-dividend-paying stocks. Consider two European call options, one on ABC and one on XYZ, both with the same time to maturity and both struck at-the-money. If the vol of ABC is greater than the vol of XYZ, is the value of the call on ABC always greater than the value of the call on XYZ?

12. In Section A.1, we discussed that an increase in time to expiration does not necessarily increase the value of European options. For example, we considered the case of a European call option on a stock that expires in a month. If the company is planning to issue a liquidating dividend, say, the day after expiry, this option is probably going to be worth more than an otherwise similar European call option that expires in, say, two months. We also considered the case of a European put option on a stock whose current price was zero. In this example an increase in time to expiration only serves to delay the time until you obtain the strike and thereby decreases the value of the put.

Consider now a non-dividend-paying asset and denote by $c(X, t, T)$ and $p(X, t, T)$ the value at time t of a call and put option, respectively, on the underlying asset that has a strike of X and expires at time T.

a) Consider the two quantities $c(Xe^{-r(T-t)}, 0, t)$ and $c(X, 0, T)$. Can you definitively say that one of these must be worth at least as much as the other? If so, show why. If not, explain why not.

b) Consider the two quantities $p(Xe^{-r(T-t)}, 0, t)$ and $p(X, 0, T)$. Can you definitively say that one of these must be worth at least as much as the other? If so, show why. If not, explain why not.

c) How do you interpret these findings?

13. In Section A.4, we derived the value of a call option in a one-period binomial model in two distinct ways, one in accordance with setting up a hedge portfolio and one in accordance with risk-neutral valuation. The resul-

tant valuation formulas were given in Equations A.14 and A.15, respectively. Show that these two valuation formulas are equivalent.

14. Consider a non-dividend-paying asset and consider a call and a put on the asset, both with the same strike and time to expiration. Denote by t today, T the expiry of the option, and r the continuously compounded risk-free rate. Suppose we use Black-Scholes to price options.

a) If the common strike on the two options is set to S, the current asset price, which of the two options is worth more?

b) If the common strike on the two options is set to $Se^{r(T-t)}$, which of the two options is worth more?

15. Assume that the standard Black-Scholes assumptions hold. Consider a non-dividend-paying stock whose current price is \$50. Assume that the annualized volatility of the stock is 20% and that the continuously compounded risk-free rate is 5%.

a) What is the value today of an option that pays off \$1mm if the stock price in three months is below a strike $X_2 = \$48$ and nothing otherwise?

b) What is the value today of an option that pays off \$1mm if the stock price in three months is less than $X_2 = \$48$ and greater than $X_1 = \$45$ and nothing otherwise? This type of option is known as a *binary corridor*. Explain why the corridor is worth less than the cash-or-nothing put option from part a).

c) Compute numerically and plot the vega for the binary corridor over a sensible range of stock prices.

d) Compute numerically and plot the delta for the binary corridor over a sensible range of stock prices. Do the same thing for time to expiration of one month, one week, and one day.

16. A *chooser option* is a European option that matures at time T, but the buyer does not have to determine whether the option is a put or a call at the time of purchase. The buyer has until time t to make this choice, where $t < T$. Assume that the call and the put both have the same strike price and time to expiry, and we will assume that the underlying asset pays no dividends.

The value of a chooser option at any time after the choice has been made (i.e., between times t and T) is just the value of a European call or put, depending on the choice made by the holder.

Define $c(X,t,T)$ as the value of a European call option at time t with a strike of X that expires at time T. Notice that the time remaining until maturity is $T-t$. Define $p(X,t,T)$ analogously.

At time t, the owner of the chooser will rationally choose whichever of the call or put has greater value. Thus, at time t, the value of the chooser is

$$\max\left[c(X,t,T), p(X,t,T)\right]. \tag{A.60}$$

a) Show that the value of the chooser today is given by

$$c(X,0,T) + p(Xe^{-r(T-t)},0,t). \tag{A.61}$$

Hint: Start by using put-call parity, which says that

$$p(X,t,T) = c(X,t,T) - S + Xe^{-r(T-t)},$$

to substitute into the expression for the put in Equation A.60.

b) "Go the other way around" and substitute in for the value of a call in Equation A.60 using put-call parity as given in the form

$$c(X,t,T) = p(X,t,T) + S - Xe^{-r(T-t)}$$

and show in this manner that the value of the chooser option must be equal to

$$p(X,0,T) + c(Xe^{-r(T-t)},0,t). \tag{A.62}$$

c) We have now derived two expressions for the value today of the chooser option, one as in Equation A.61 and the other as in Equation A.62. Show that these two formulations are equivalent.

d) A *straddle* is a position that is long both a call and a put with the same strike and the same expiry. The value of a straddle that matures at time T is

$$c(X,0,T) + p(X,0,T). \tag{A.63}$$